Target statement: IMO 2024 P1, with answer pre-populated by Gemini 1.5 Pro

```
: {α : ℝ | ∀ n > (0 : ℤ), n | ∑ k in Finset.Icc (1 : ℤ) n, ⌊k * α⌋}
  = {α : ℝ | ∃ k : ℤ, Even k ∧ α = k}
```



Variant generator

Simplification

```
: {α : ℚ | ∀ n > (0 : ℤ), n | ∑ k in Finset.Icc (1 : ℤ) n, ⌊k * α⌋}
  = {α : ℚ | ∃ a : ℤ, α = a ∧ Even a}
```

a Consider a statement for rational ▯ only

Simplification

```
: {(α : ℝ) | ∀ (n : ℕ), 0 < n → (n : ℤ) | (∑ i in Finset.Icc 1 n, ⌊i * α⌋) ∧
    (n : ℤ) | (∑ i in Finset.Icc 1 n, ⌊(1 + i) * α⌋)}
  = {α : ℝ | ∃ k : ℤ, α = 2 * k}
```

b Assume a stronger property holds for α

Lemma

```
: ∃ (α : ℝ), ∀ n > 3, n | ∑ k in Finset.Icc (1 : ℤ) n, ⌊k * α⌋ ∧ 1 < α ∧ α < 2
```

c There is an α with $1 < \alpha < 2$ (disprovable)

Lemma

```
{α : ℝ} (hα : ∀ (n : ℕ), 0 < n → (n : ℤ) | (∑ i in Finset.Icc 1 n, ⌊i * α⌋)) :
  ∃ k : ℤ, |α - k| < 1 / 4
```

d There is an integer within distance ¼ of α

Proof step

```
: {α : ℝ | ∀ n > (0 : ℤ), n | ∑ k in Finset.Icc (1 : ℤ) n, ⌊k * α⌋}
  ⊆ {α : ℝ | ∃ q : ℚ, α = q ∧ α = ⌊α⌋}
```

e Statement effectively claiming that α has to be an integer

Proof step

```
: {α : ℝ | ∃ q : ℚ, α = q ∧ ∀ n > (0 : ℤ), n | ∑ k in Finset.Icc (1 : ℤ) n, ⌊k * α⌋}
  = {α : ℝ | ∃ q : ℚ, α = q ∧ ∀ n > (5 : ℤ), n | ∑ k in Finset.Icc (1 : ℤ) n, ⌊k * α⌋}
```

f Statement effectively claiming that α has to be a rational number

Reformulation

```
: {α : ℝ | ∀ n > (0 : ℤ), n | ∑ k in Finset.Icc (1 : ℤ) n, ⌊α * k⌋}
  = {α : ℝ | ∀ n > (0 : ℤ), n | ∑ k in Finset.Icc (1 : ℤ) n, ⌈α * k⌉}
```

g Considering ceiling formulation

Reformulation

```
: {(α : ℝ) | ∀ (n : ℕ), 0 < n → (n : ℤ) | (- (∑ i in Finset.Icc 1 n, ⌊i * α⌋))}
  = {α : ℝ | ∃ k : ℤ, Even k ∧ α = k}
```

h Replace sum by its negative

Analogous statement

```
: {α : ℝ | ∃ k : ℤ, α = 2 * k}
  ⊂ {α : ℝ | ∀ (n : ℕ), 0 < n → (n : ℤ) | (∑ i in Finset.Icc 1 n, ⌊i * α⌋ + ⌊n * α⌋)}
```

i Add $\lfloor n\alpha \rfloor$ and replace equality by $\subset$ (disprovable)

Analogous statement

```
: {(α : ℝ) | ∀ (n : ℕ), 0 < n → (n : ℤ) | (∑ i in Finset.Icc 1 n, ⌊i * α + 0.6⌋)}
  = {2 * k | k ∈ Set.range (Int.cast : ℤ → ℝ)}
```

j Add 0.6 to the value in floor